

HTTP methods

HTTP methods define what action the client want to perform on the server.

Think of them like action in restaurant

Action	HTTP method
see menu	Get
place order	Post
update order	PUT
modify item	PATCH
cancel order	DELETE

What are HTTP methods?

HTTP methods are verbs used in API request to tell the server: "What do you want me to do with this resources?"

Example -

GET / users

Post / users

PUT / users / 1

DELETE / users / 1





1 GET - Fetch Data

used to retrieve data from server.

Characteristics:

No data modification

Safe

Cacheable

can be bookmarked

Example:

GET /users

Real life example:

```
@app.get("/users")
```

```
def get_users():
```

```
    return users
```

2. Post - Create New Data

used to send data and create new resources.

Characteristics:

changes server data

Not idempotent

Data sent in request body

Example: POST /users

Real-life example:

Filling a registration form

```
@app.post("/users")
```

```
def create_user(user: User):
```

```
    return user
```

3. PUT - Update Entire Resource

Replace entire object



Characteristics :

Idempotent

Explain idempotent

1. If you send a PUT request once to set a resource to a specific state (eg. updating a user's email address to "test@example.com"), the server's data will be updated. If you send that exact same PUT request five more times, the resource's state will not change further, it will remain "test@example.com".

2. Reliability for Retries

This characteristic is crucial in unreliable network environment. If a client sends a PUT request but doesn't receive a response due to network timeout. It can safely retry the same request, knowing it won't corrupt data or cause duplicate entries. The server effectively processes the first request and ignores subsequent identical ones, ensuring the final state is consistent.

Example :

PUT /users/1



Real life:

Page No. 15

Rewriting a document completely

Example:

```
@app.put("/users/{id}")
def update_user(id: int, user: User):
    return user
```

(4) PATCH - Update Partial Data

Update only specific fields

Difference from PUT:

PUT

PATCH

Full Update

Partial Update

Replace

Modify

Example:

am PATCH /users/1

Example -

```
@app.patch("/users/{id}")
def update_email(id: int, email: str):
    return {"id": id, "email": email}
```

(5) DELETE - Remove Data

Delete a resource

Example -

```
@app.delete("/users/{id}")
def delete_user(id: int):
    return {"message": "user deleted"}
```